PHYSICS • GRADE 8

Detailed study notes

Thermal, electrical, magnetic and optical phenomena

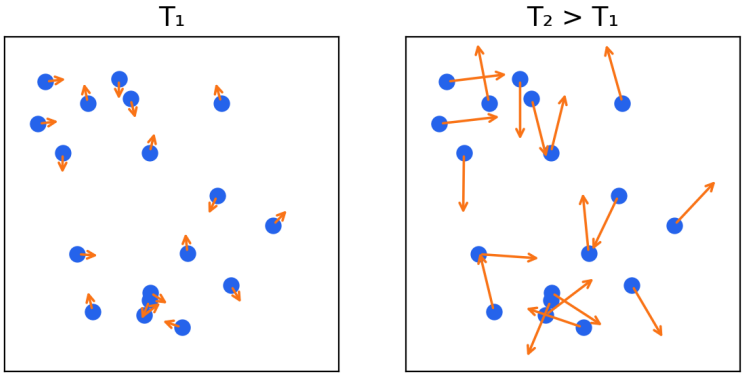
Based on the textbook: N. A. Zakirova & R. R. Ashirov. Physics. Grade 8. — Almaty: Arman-PV, 2018.

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Chapter 1. Thermal phenomena



§ 1. Thermal motion, Brownian motion and diffusion

- Matter is made of particles that are continuously moving and interacting; their random motion is thermal motion.
- Brownian motion is visible evidence of molecular motion: suspended fine particles move irregularly because molecules strike them unevenly.
- Diffusion is the mutual penetration of particles of different substances; it is faster in gases, slower in liquids and very slow in solids.

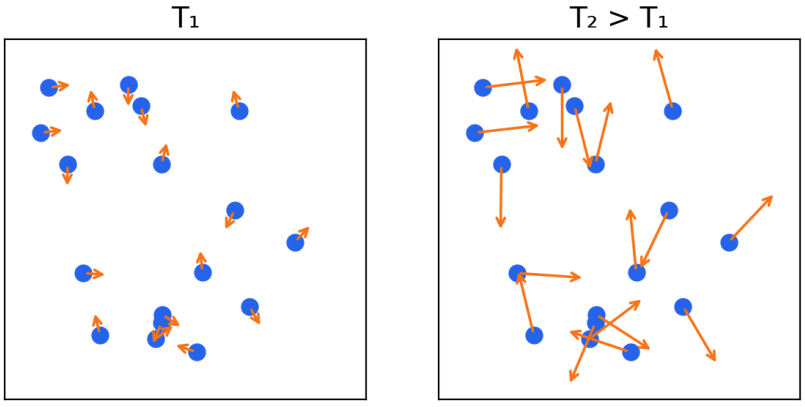
DEFINITIONS

Term	Definition
Thermal motion	random, continuous motion of molecules and atoms that make up a body.
Brownian motion	continuous random motion of tiny solid particles suspended in a liquid or gas, caused by molecular impacts.
Diffusion	spontaneous interpenetration of particles of different substances.

FORMULAE AND RELATIONS

Formula	Note
T ↑ → v ↑	heating increases the average speed of particles and makes diffusion faster.

DIAGRAM §1



KEY POINT
The smell of perfume spreading through a room is diffusion in a gas; it does not stop even at negative temperatures.

§ 2. Temperature, measurement methods and temperature scales

- Temperature characterizes the degree of heating of a body and is related to the average kinetic energy of its particles.
- A thermometer uses a property that changes predictably with temperature, most often thermal expansion of a liquid or gas.
- Celsius, Kelvin and Fahrenheit scales use different reference points. Kelvin is an absolute scale; 0 K corresponds to −273.15 °C.

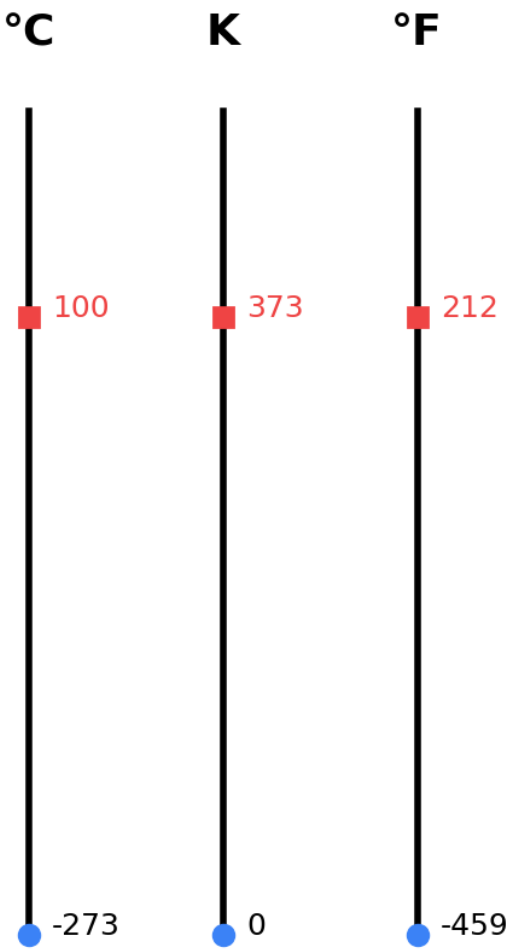
DEFINITIONS

Term	Definition
Temperature	a physical quantity that characterizes the thermal state (degree of heating) of a body.
Thermometer	an instrument for measuring temperature.
Absolute zero	the lowest possible temperature: 0 K = −273.15 °C.

FORMULAE AND RELATIONS

Formula	Note
T = t + 273	conversion from Celsius t (°C) to Kelvin T (K), approximately.
t°F = 1.8·t°C + 32	conversion from Celsius to Fahrenheit.

DIAGRAM §2



KEY POINT
A thermometer must be in thermal contact with the body long enough for their temperatures to become equal.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 1, item 3

Convert 40 °C to °F.

Given	SI	Solution
t = 40 °C t_F = ?	*No conversion to SI required*	t_F = 1.8·t_C + 32 t_F = 1.8·40 + 32 t_F = 104 °F

Answer: 40 °C = 104 °F.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 3. Internal energy and ways of changing it

- A body's internal energy is the sum of the kinetic energies of random particle motion and the potential energies of their interaction.
- Internal energy changes in two ways: by doing mechanical work and by heat transfer.

- When work is done on a gas (compression), its internal energy increases; when a gas expands and does work, its internal energy decreases.

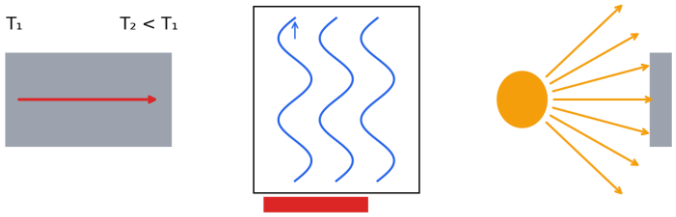
DEFINITIONS

Term	Definition
Internal energy U	the sum of the kinetic energy of thermal motion and potential energy of interaction of a body's particles.
Heat transfer	change of internal energy without mechanical work, due to energy transfer between bodies.

FORMULAE AND RELATIONS

Formula	Note
ΔU = Q + A	general energy statement: internal energy changes because of heat Q supplied to the body and work A done on it.
[U] = J	the SI unit of internal energy is the joule.

DIAGRAM §3



KEY POINT
Friction, impact, compression and stirring can increase internal energy because mechanical work becomes microscopic motion.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 2, item 3

How does the internal energy of a body change if 15 J of work is done on it?

Given	SI	Solution
A = 15 J ΔU = ?	*All values are in SI; Q = 0*	ΔU = Q + A ΔU = 0 + 15 J = 15 J

Answer: The internal energy increases by 15 J.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 4. Conduction, convection and radiation

- Heat conduction transfers energy through a substance without bulk transfer of matter; metals are generally good conductors.
- Convection transfers energy by motion of layers of a liquid or gas; it cannot occur in a solid or in vacuum.
- Thermal radiation transfers energy by electromagnetic waves and can propagate through vacuum. Dark, dull surfaces absorb and emit radiation well.

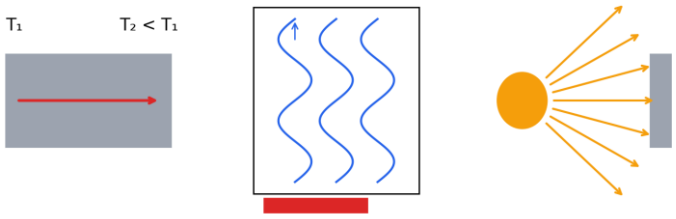
DEFINITIONS

Term	Definition
Conduction	transfer of internal energy from hotter parts to colder parts of a body without transfer of matter.
Convection	heat transfer by streams of a liquid or gas.
Thermal radiation	heat transfer by electromagnetic radiation.

FORMULAE AND RELATIONS

Formula	Note
Q_cond: hot → cold	conduction continues while a temperature difference exists.
Q_rad ∝ surface properties	absorption and emission depend strongly on colour and surface finish.

DIAGRAM §4



KEY POINT
A paper container can boil water because heat is conducted into the water; while water is present, the paper stays near 100 °C.

§ 5. Heat transfer in nature and technology

- Natural and engineered systems use conduction, convection and radiation together; the dominant process depends on material and environment.
- Thermal insulation reduces unwanted energy transfer. Porous materials, fur, wool and trapped air are poor heat conductors.
- Thermos flasks use a double wall with a vacuum, reflective surfaces and a stopper to reduce all three forms of heat transfer.

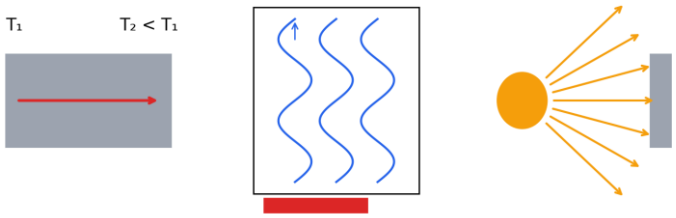
DEFINITIONS

Term	Definition
Thermal insulator	a material or structure that weakly conducts heat and reduces heat loss or gain.
Forced convection	convection caused by a fan, pump or another device.

FORMULAE AND RELATIONS

Formula	Note
thermal resistance ↑ → heat loss ↓	thicker insulating layers and trapped air reduce heat transfer.

DIAGRAM §5



KEY POINT
Insulation is not "warm"; it slows the transfer of energy, so it helps keep warm things warm and cold things cold.

§ 6. The role of thermal phenomena in living organisms

- Living organisms exchange energy with their environment by conduction, convection, radiation and evaporation.
- Endothermic animals maintain a nearly constant body temperature through metabolism and thermoregulation; ectothermic animals depend more on environmental temperature.
- Adaptations such as fur, feathers, fat layers, sweating, panting and burrowing control heat loss and heat gain.

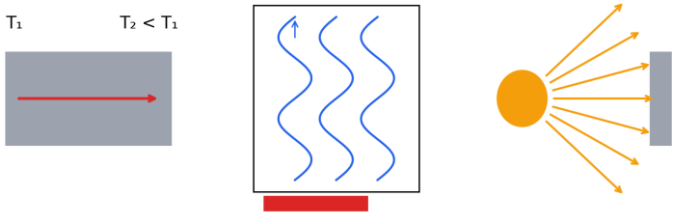
DEFINITIONS

Term	Definition
Thermoregulation	maintenance of a suitable body temperature by an organism.
Evaporation cooling	lowering of temperature because the fastest molecules escape from a liquid surface.

FORMULAE AND RELATIONS

Formula	Note
evaporation ↑ → cooling ↑	sweat evaporation removes energy from the skin.

DIAGRAM §6



KEY POINT
In hot, dry air evaporation is effective; in humid air sweat evaporates more slowly and cooling is weaker.

§ 7. Quantity of heat and specific heat capacity

- The quantity of heat is energy received or released by a body during heat transfer.
- For the same mass and temperature change, different substances require different quantities of heat; this is described by specific heat capacity.

- Water has a high specific heat capacity, so large bodies of water heat and cool slowly.

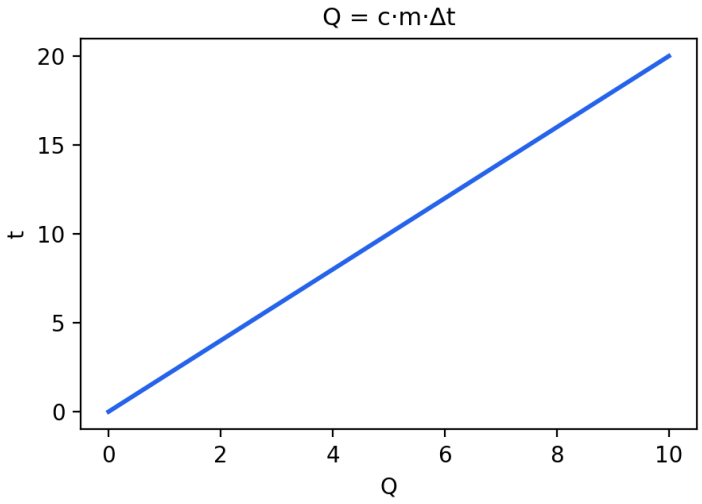
DEFINITIONS

Term	Definition
Quantity of heat Q	energy transferred during heat exchange.
Specific heat capacity c	the heat required to raise the temperature of 1 kg of a substance by 1 °C (or 1 K).
Heat capacity C	heat required to change the temperature of a given body by 1 °C.

FORMULAE AND RELATIONS

Formula	Note
$Q = c \cdot m \cdot (t_2 - t_1)$	heat gained on heating; Q is positive when $t_2 > t_1$.
$C = c \cdot m$	heat capacity of a body.

DIAGRAM §7



KEY POINT
Use kilograms, joules per kilogram-degree and degrees Celsius (or kelvins) consistently in calculations.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 5, item 2

An aluminium spoon of mass 50 g at 25 °C is put into hot water at 75 °C. What heat does the spoon receive?

Given	SI	Solution
m = 50 g t_1 = 25 °C t_2 = 75 °C c = 920 J/(kg·°C)	*m = 0.050 kg*	$Q = c \cdot m \cdot (t_2 - t_1)$ $Q = 920 \cdot 0.050 \cdot (75 - 25)$ Q = 2300 J

Answer: Q = 2.3 kJ.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 8. Fuel energy and specific heat of combustion

- Fuel contains chemical energy. During complete combustion, chemical energy is released mainly as internal energy of hot products.
- The specific heat of combustion tells how much heat is released when 1 kg of fuel burns completely.
- Practical fuel choice also depends on availability, cost, safety, transport and environmental impact.

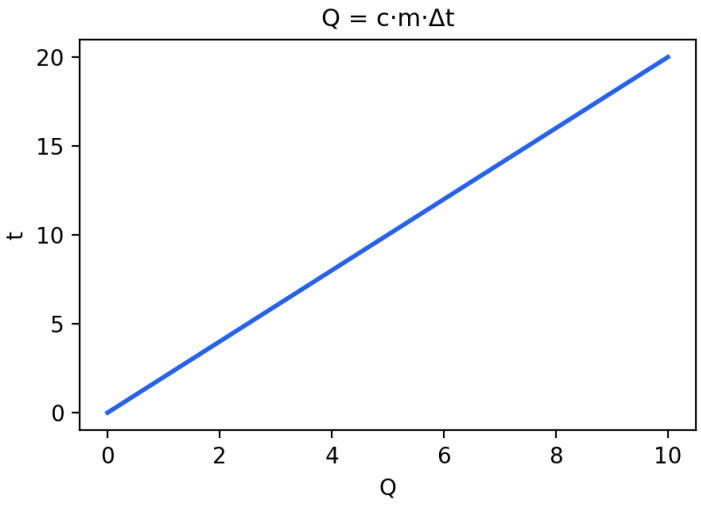
DEFINITIONS

Term	Definition
Complete combustion	burning of fuel in sufficient oxygen with maximum possible heat release.
Specific heat of combustion q	heat released by complete combustion of 1 kg of fuel.

FORMULAE AND RELATIONS

Formula	Note
Q = q·m	heat released by complete combustion of fuel mass m.
[q] = J/kg	SI unit of specific heat of combustion.

DIAGRAM §8



KEY POINT
The textbook table gives q = 30 MJ/kg for hard coal and q = 42.7 MJ/kg for diesel fuel.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 6, item 1

How much energy is released by complete combustion of 15 kg of hard coal?

Given	SI	Solution
m = 15 kg q = 30 MJ/kg	*All values use compatible units*	Q = q·m Q = 30 MJ/kg · 15 kg Q = 450 MJ

Answer: Q = 450 MJ.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 9. Conservation and transformation of energy in thermal processes

- Energy is neither created nor destroyed; in thermal processes it is transferred and transformed between mechanical, internal and other forms.
- When bodies at different temperatures exchange heat in an isolated system, energy lost by hotter bodies equals energy gained by colder bodies.
- The heat-balance equation is used for mixtures and calorimetry, after accounting for all bodies that exchange heat.

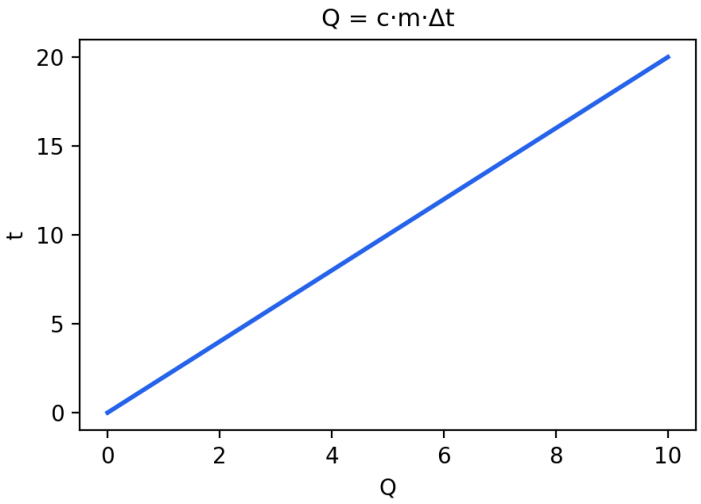
DEFINITIONS

Term	Definition
Heat balance	the equality of total heat given off and total heat absorbed in an isolated heat-exchange system.
Calorimeter	a device designed to reduce heat exchange with the surroundings when studying thermal processes.

FORMULAE AND RELATIONS

Formula	Note
ΣQ = 0	algebraic sum of heats in an isolated system; heat lost is taken with a minus sign.
Q_hot = Q_cold	simplified heat balance for two bodies when losses are neglected.

DIAGRAM §9



KEY POINT
For mixtures of water, mass can be found from volume using ρwater ≈ 1000 kg/m³, so 1 L of water has a mass close to 1 kg.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 7, item 1

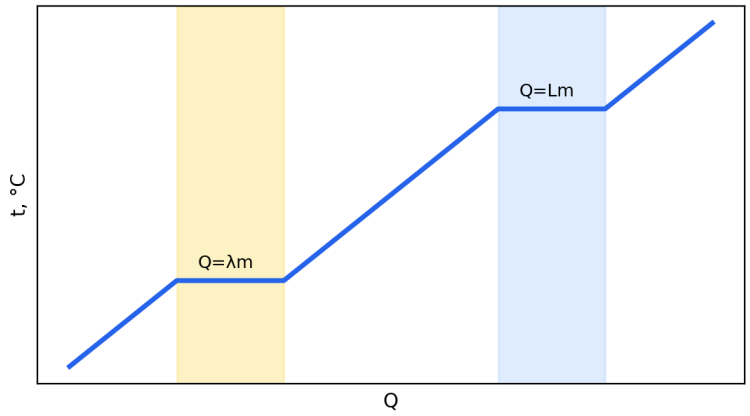
To prepare bath water at 40 °C, cold water at 10 °C is mixed with hot water at 60 °C. What volume of hot water is added to 80 L of cold water?

Given	SI	Solution
$V_1 = 80$ л $t_1 = 10$ °C $t_2 = 60$ °C $t = 40$ °C	*For water: $m \propto V$*	$m_2 \cdot c \cdot (60-40) = m_1 \cdot c \cdot (40-10)$ $V_2 \cdot 20 = 80 \cdot 30$ $V_2 = 120$ л

Answer: 120 L of hot water.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

Chapter 2. States of matter



§ 10. Melting and crystallization of solids; melting point and specific latent heat of fusion

- Melting is the transition from solid to liquid; crystallization (solidification) is the reverse process.
- For a crystalline substance at constant pressure, melting and crystallization occur at a definite temperature. During the phase change, supplied or released energy changes interactions rather than temperature.
- The specific latent heat of fusion depends on the substance and gives the heat needed to melt 1 kg at its melting point.

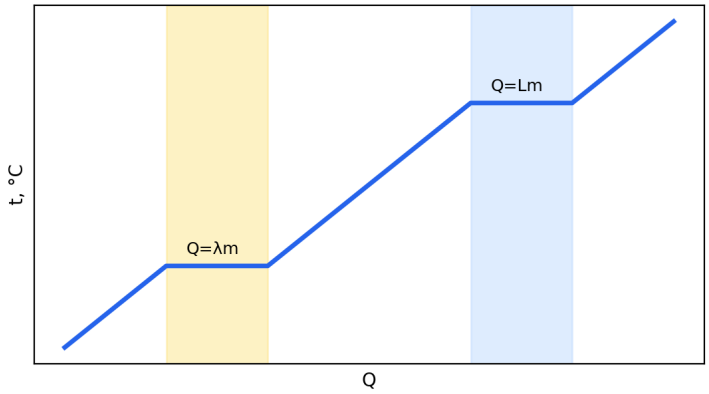
DEFINITIONS

Term	Definition
Melting	transition of a substance from solid state to liquid state.
Crystallization	transition from liquid state to solid state.
Melting point	temperature at which a crystalline substance melts (and solidifies) at a given pressure.

FORMULAE AND RELATIONS

Formula	Note
$Q = \lambda \cdot m$	heat needed to melt mass m at the melting point; the same magnitude is released on crystallization.
$Q = c \cdot m \cdot \Delta t + \lambda \cdot m$	for heating a solid to its melting point and then melting it.

DIAGRAM §10



KEY POINT
The textbook table gives λ = 2.7×10⁵ J/kg and λice = 3.4×10⁵ J/kg.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 8, item 1

Find the energy required to turn 100 kg of iron at its melting temperature into liquid.

Given	SI	Solution
m = 100 kg λ = 2.7×10⁵ J/kg	*All values are in SI*	Q = λm Q = 2.7×10⁵ · 100 Q = 2.7×10⁷ J

Answer: Q = 27 MJ.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 11. Vaporization and condensation

- Vaporization is the transition from liquid to gas. It can occur at the surface at any temperature (evaporation) or throughout a liquid at its boiling point.
- Evaporation is faster when temperature and free surface area are larger, air above the liquid is drier, and vapour is removed by wind.
- Condensation is the transition from gas to liquid and is accompanied by release of energy.

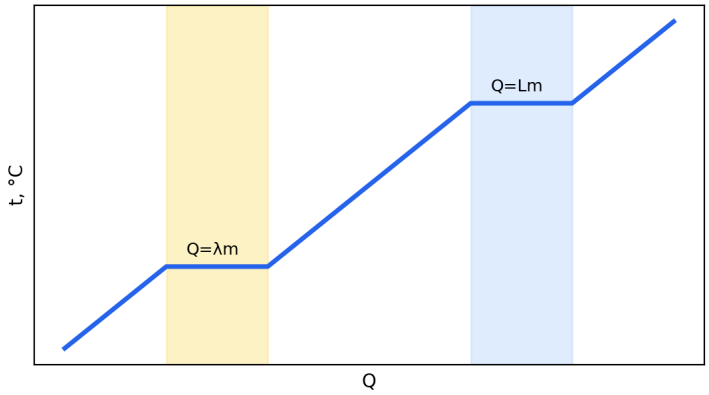
DEFINITIONS

Term	Definition
Evaporation	vaporization occurring from the free surface of a liquid at any temperature.
Condensation	transition of a vapour or gas to a liquid.
Saturated vapour	vapour in dynamic equilibrium with its liquid in a closed space at a given temperature.

FORMULAE AND RELATIONS

Formula	Note
evaporation → U decreases	the fastest molecules leave, lowering the average kinetic energy of the remaining liquid.
condensation → energy released	forming a liquid strengthens molecular interactions.

DIAGRAM §11



KEY POINT
Wet clothes dry faster in moving, dry and warm air because all three conditions make evaporation faster.

§ 12. Boiling, specific latent heat of vaporization and the dependence of boiling point on external pressure

- Boiling is vaporization throughout the liquid with formation and growth of vapour bubbles. It begins when saturated-vapour pressure becomes equal to external pressure.
- At a fixed external pressure a pure liquid boils at a definite temperature. Increasing external pressure raises the boiling point; decreasing it lowers the boiling point.
- The specific latent heat of vaporization is the heat needed to turn 1 kg of liquid at its boiling point into vapour at the same temperature.

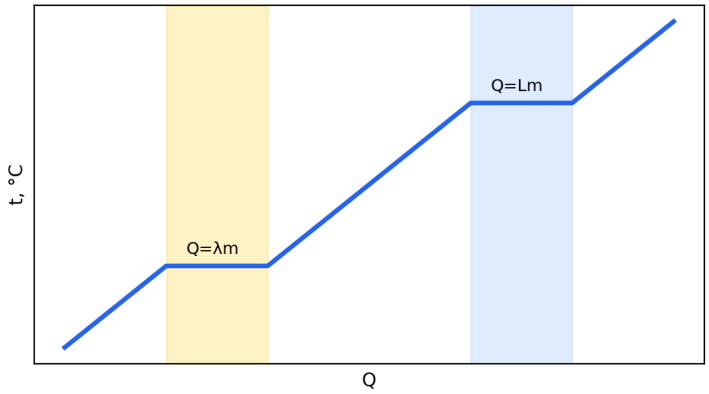
DEFINITIONS

Term	Definition
Boiling	vaporization occurring throughout the volume of a liquid with bubble formation.
Specific latent heat of vaporization L	heat needed to vaporize 1 kg of a liquid at its boiling point.
Boiling point	temperature at which a liquid boils at a specified external pressure.

FORMULAE AND RELATIONS

Formula	Note
$Q = L \cdot m$	heat required for vaporization at the boiling point.
$Q_{total} = c \cdot m \cdot (t_b - t_1) + L \cdot m$	for heating a liquid to boiling and vaporizing it completely.

DIAGRAM §12



KEY POINT
A pressure cooker raises pressure and hence boiling temperature, which speeds up cooking; at high altitude water boils below 100 °C.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 9, item 3

360 kJ of energy is used to vaporize alcohol at its boiling point. Find the mass of alcohol vaporized.

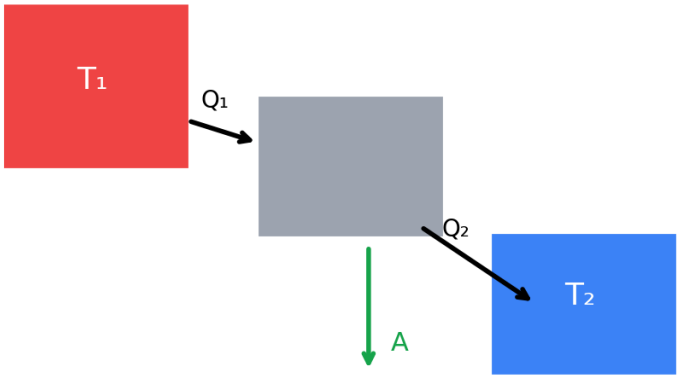
Given	SI	Solution
Q = 360 kJ L = 0.9 MJ/kg	*Q = 0.360 MJ*	Q = Lm m = Q/L m = 0.360/0.9 = 0.40 kg

Answer: m = 0.40 kg.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

Chapter 3. Fundamentals of thermodynamics

$$Q_1 = A + Q_2, \quad \eta = A/Q_1$$



§ 13. The first law of thermodynamics; work done by gas and vapour

- The first law of thermodynamics expresses energy conservation for thermal systems.
- A gas does positive work when it expands against external pressure; external forces do work on the gas when it is compressed.
- At constant pressure the work of a gas is proportional to its change in volume.

DEFINITIONS

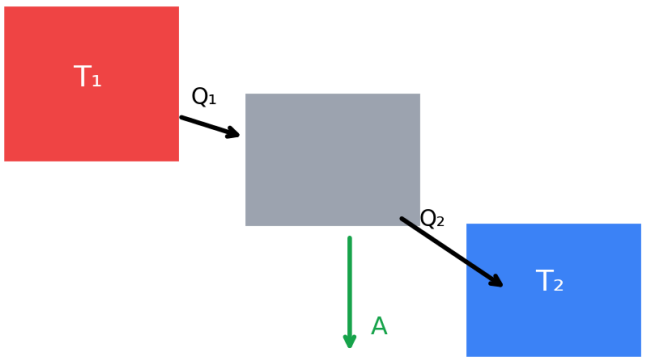
Term	Definition
Thermodynamic system	a body or group of bodies considered when studying energy exchange.
Work of a gas	energy transferred by a gas as it changes volume against external pressure.

FORMULAE AND RELATIONS

Formula	Note
ΔU = Q − A_gas	first law when A_gas is work done by the gas: supplied heat changes internal energy and/or becomes work.
A_gas = p·ΔV	work in an isobaric (constant-pressure) process.

DIAGRAM §13

$$Q_1 = A + Q_2, \quad \eta = A/Q_1$$



KEY POINT
Be explicit about the sign convention: work done by the gas and work done on the gas have equal magnitudes but opposite signs.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 10, item 1

What work is done by a gas expanding at constant pressure 2 MPa from 2 L to 12 L?

Given	SI	Solution
p = 2 MPa V_1 = 2 л V_2 = 12 л	*p = 2×10⁶ Pa; ΔV = 10×10⁻³ m³*	A = pΔV A = 2×10⁶ · 10×10⁻³ A = 2.0×10⁴ J

Answer: A = 20 kJ.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 14. Irreversibility of thermal processes and the second law of thermodynamics

- Many natural thermal processes have a preferred direction: heat flows spontaneously from hot to cold, gases diffuse, and friction converts ordered motion into internal energy.
- A process is irreversible if neither the system nor the surroundings can both be restored exactly to their original states without other changes.
- The second law rules out a cyclic heat engine that converts all heat received from one reservoir entirely into work.

DEFINITIONS

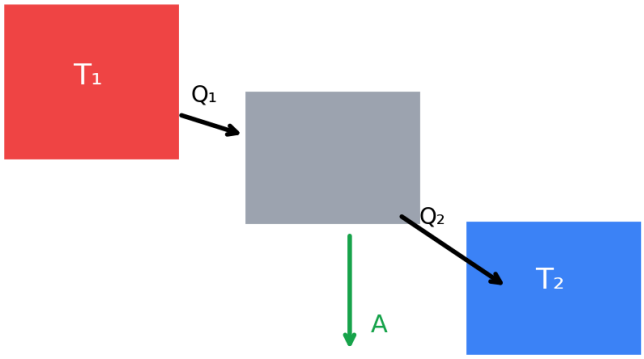
Term	Definition
Irreversible process	a process that cannot be reversed so that both system and surroundings return exactly to their initial states.
Second law of thermodynamics	a law that specifies the direction of spontaneous thermal processes and limits heat-engine efficiency.

FORMULAE AND RELATIONS

Formula	Note
heat: T_hot → T_cold	spontaneous heat flow requires a temperature difference and proceeds from hotter to colder.
η < 100%	no real or cyclic heat engine can convert all input heat to useful work.

DIAGRAM §14

$$Q_1 = A + Q_2, \quad \eta = A/Q_1$$



KEY POINT
A movie of a warm cup cooling looks physically natural; the reverse movie—a cup becoming hotter while the room cools—does not happen spontaneously.

§ 15. Heat engines

- A heat engine operates cyclically: it receives heat from a heater, converts part into work and transfers the rest to a cooler.
- External-combustion engines heat a working substance outside the engine; internal-combustion engines burn fuel inside their cylinders.
- Steam turbines, internal-combustion engines and jet engines convert thermal energy to mechanical energy in different ways.

DEFINITIONS

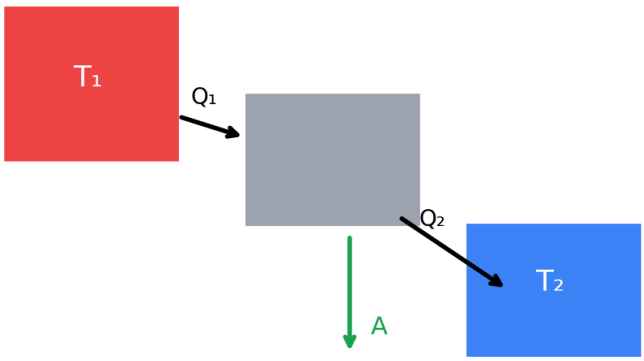
Term	Definition
Heat engine	a cyclic device that converts part of heat received from a heater into mechanical work.
Heater	a body or source at higher temperature that gives heat to the working substance.
Cooler	a body at lower temperature that receives unused heat.

FORMULAE AND RELATIONS

Formula	Note
$Q_1 = A + Q_2$	energy balance for one engine cycle: heat received equals useful work plus heat rejected.

DIAGRAM §15

$$Q_1 = A + Q_2, \quad \eta = A/Q_1$$



KEY POINT
The cooler is not an optional part: it is required because some energy must be rejected in every real heat-engine cycle.

§ 16. Efficiency of a heat engine

- Efficiency compares useful work with the heat supplied by the heater. It is usually given as a percentage.
- Because heat is rejected to a cooler and because of friction and other losses, efficiency is always below 100%.
- For an ideal reversible engine, efficiency depends on the absolute temperatures of heater and cooler.

DEFINITIONS

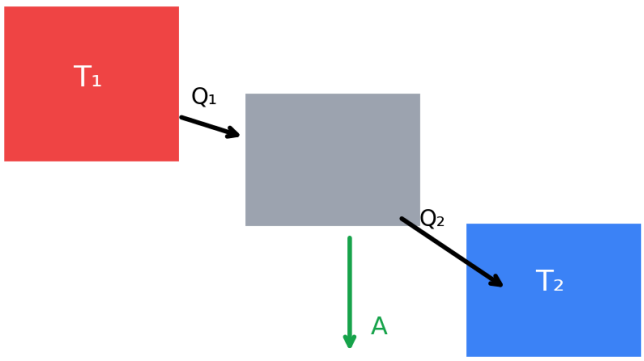
Term	Definition
Efficiency η	ratio of useful work to heat received by a heat engine.
Useful work	the mechanical work delivered by an engine for its intended purpose.

FORMULAE AND RELATIONS

Formula	Note
$\eta = A/Q_1 \cdot 100\%$	efficiency from useful work A and heat Q_1 received from heater.
$\eta = (1 - Q_2/Q_1)\cdot 100\%$	equivalent form using rejected heat Q_2.
$\eta_ideal = (1 - T_2/T_1)\cdot 100\%$	ideal-engine limit; T_1 and T_2 must be in kelvins.

DIAGRAM §16

$$Q_1 = A + Q_2, \quad \eta = A/Q_1$$



KEY POINT
Always convert megajoules/kilojoules and use kelvins, not degrees Celsius, in the ideal-engine temperature formula.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 11, item 2

A heat engine has efficiency 40% and receives 5 kJ of heat from the heater. What work does it do, and how much heat is lost?

Given	SI	Solution
$\eta = 40\% = 0.40$ $Q_1 = 5$ kJ	*All values use kJ*	$\eta = A/Q_1$ $A = \eta Q_1 = 0.40 \cdot 5 = 2$ kJ $Q_2 = Q_1 - A = 5 - 2 = 3$ kJ

Answer: $A = 2$ kJ; $Q_2 = 3$ kJ.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 17. Environmental problems of using heat machines

- Combustion engines and thermal power plants release carbon dioxide, nitrogen oxides, sulfur compounds, soot and waste heat.
- The environmental impact depends on fuel composition, combustion completeness, energy demand and pollution-control technology.
- Reducing impact requires energy efficiency, cleaner fuels, emissions control, public transport and renewable energy sources.

DEFINITIONS

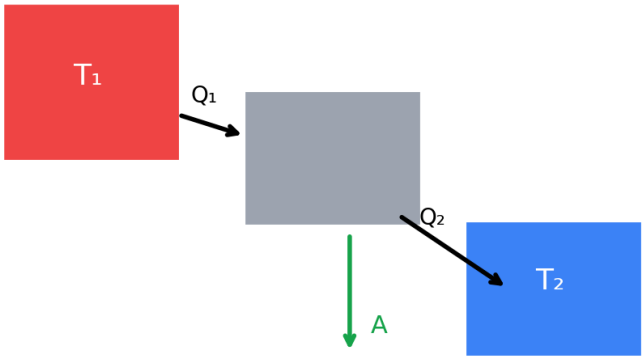
Term	Definition
Greenhouse gas	a gas that absorbs outgoing infrared radiation and contributes to the greenhouse effect.
Thermal pollution	environmental warming caused by waste heat released from technology.
Renewable energy	energy from sources replenished naturally on a human time scale.

FORMULAE AND RELATIONS

Formula	Note
fuel use ↓ → emissions ↓	improving efficiency reduces fuel consumption and, usually, pollutant emissions per useful task.

DIAGRAM §17

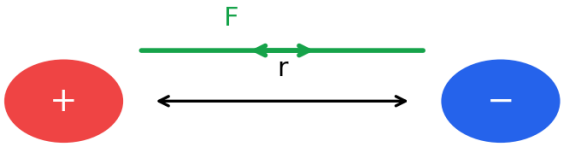
$$Q_1 = A + Q_2, \quad \eta = A/Q_1$$



KEY POINT
Energy conservation is an environmental measure: energy that is not wasted does not need to be generated.

Chapter 4. Fundamentals of electrostatics

$$F = k\cdot|q_1q_2| / (\varepsilon\cdot r^2)$$



§ 18. Electric charge, electrification, conductors and dielectrics

- Electric charge is a physical property that determines electromagnetic interactions. Charges have two signs: positive and negative.
- Electrification transfers or redistributes electrons. Like charges repel; unlike charges attract.
- Conductors contain mobile charge carriers; in dielectrics charges are bound, but they can be polarized in an electric field.

DEFINITIONS

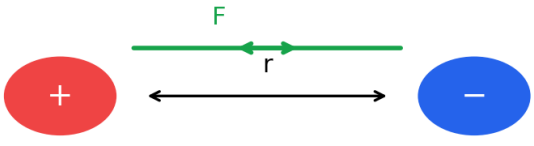
Term	Definition
Electric charge q	a physical quantity characterizing the ability of bodies or particles to take part in electric interactions.
Electrification	the process by which a body acquires an electric charge.
Conductor / dielectric	a conductor has mobile charge carriers; a dielectric normally has no freely moving charges.

FORMULAE AND RELATIONS

Formula	Note
$q = \pm N\cdot e$	body charge is an integer multiple of the elementary charge e.
$e = 1.6\times10^{-19}$ C	magnitude of the elementary charge.

DIAGRAM §18

$$F = k\cdot|q_1q_2| / (\varepsilon\cdot r^2)$$



KEY POINT
A charged object may attract a neutral object because it polarizes charges inside it; attraction alone does not prove that both objects have opposite net charges.

§ 19. Conservation of electric charge, interaction of stationary charges, Coulomb's law and elementary charge

- In an isolated system, the algebraic sum of electric charges is constant. Charge can move between bodies but is not created or destroyed in ordinary processes.
- Coulomb's law gives the magnitude of the force between two point charges. The force increases with each charge and decreases with the square of distance.
- The medium weakens the force relative to vacuum by its relative permittivity.

DEFINITIONS

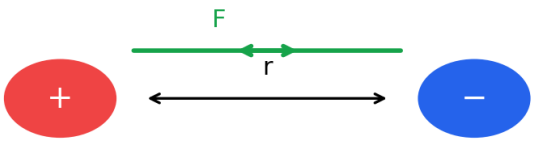
Term	Definition
Law of charge conservation	the algebraic sum of charges in an isolated system remains constant.
Point charge	a charged body whose dimensions can be neglected compared with distances in the problem.
Elementary charge	the magnitude of charge of a proton; electron charge is −e.

FORMULAE AND RELATIONS

Formula	Note
$\Sigma q = \text{const}$	law of conservation of electric charge.
$F = k\cdot\lvert q_1 q_2\rvert/(\varepsilon\cdot r^2)$	Coulomb's law in a medium; $k \approx 9.0\times10^9$ N·m²/C².

DIAGRAM §19

$$F = k\cdot|q_1 q_2| \,/\, (\varepsilon\cdot r^2)$$



KEY POINT
If the distance between point charges is doubled, the Coulomb force becomes four times smaller.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 13, item 1

Two identical conducting spheres with charges 2 nC and 8 nC are brought into contact and separated again. Find the charges after contact.

Given	SI	Solution
$q_1 = 2$ nC $q_2 = 8$ nC	*Identical spheres: equal final charges*	$q\Sigma = q_1 + q_2 = 10$ nC $q_1' = q_2' = q\Sigma/2$ $q_1' = q_2' = 5$ nC

Answer: Each sphere has +5 nC.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 20. Electric field and electric-field strength

- An electric field is a form of matter through which charges exert forces on one another. It exists around charged bodies.
- The field-strength vector gives the force per unit positive test charge; its direction is the direction of force on a positive test charge.
- Field lines begin on positive charges and end on negative charges. Where lines are denser, the field is stronger.

DEFINITIONS

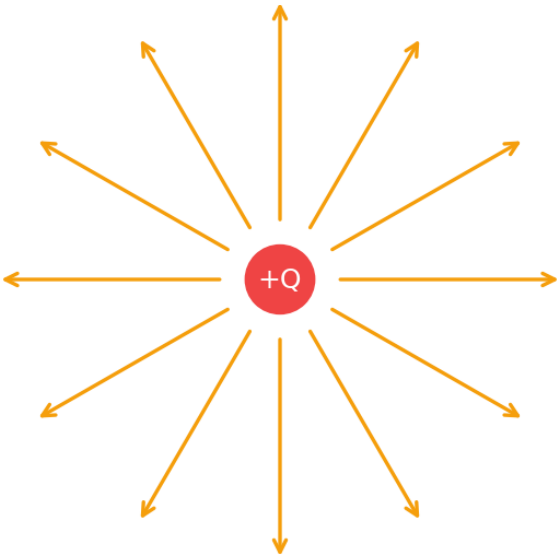
Term	Definition
Electric field	a form of matter that transmits electric interaction between charges.
Electric-field strength E	force acting on a positive test charge divided by the charge.

FORMULAE AND RELATIONS

Formula	Note
$E = F/q$	definition of electric-field strength; [E] = N/C = V/m.
$E = k\cdot\lvert Q\rvert/(\varepsilon\cdot r^2)$	strength of a point charge field in a medium.

DIAGRAM §20

$$E = F/q$$



KEY POINT
Electric field lines never cross: if they crossed, the field would have two different directions at one point.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 14, item 1

What is the electric-field strength 1 m from a point charge 0.1 nC? What force acts there on a charge 10 nC?

Given	SI	Solution
Q = 0.1 nC r = 1 m q = 10 nC	$Q = 1\times10^{-10}$ C; $q = 1\times10^{-8}$ C	$E = kQ/r^2 = 9\times10^9\cdot10^{-10}/1^2 = 0.9$ H/C $F = Eq = 0.9\cdot10^{-8} = 9\times10^{-9}$ N

Answer: E = 0.9 N/C; F = 9×10^{-9} N.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 21. Potential, potential difference and capacitor

- Potential is a scalar energy characteristic of an electric field at a point. Potential difference (voltage) is work per unit charge between two points.
- Equipotential surfaces have the same potential; moving a charge along them requires no work by the electrostatic field.
- A capacitor consists of two conductors separated by a dielectric and stores equal and opposite charges. Its capacitance measures charge stored per voltage.

DEFINITIONS

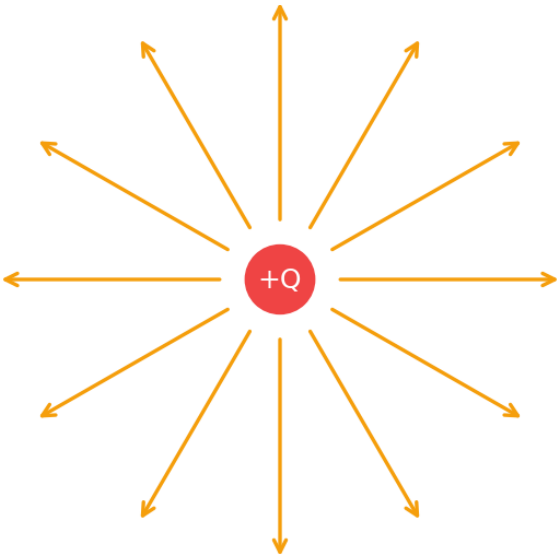
Term	Definition
Potential φ	potential energy per unit positive charge at a point of an electric field.
Voltage U	potential difference between two points; work per unit charge.
Capacitance C	ratio of a capacitor's charge to its voltage.

FORMULAE AND RELATIONS

Formula	Note
$\varphi = A/q$	potential expressed through work A per charge q.
$U = \varphi_1 - \varphi_2 = A/q$	voltage and work relation.
$C = q/U$; $C = \varepsilon\varepsilon_0 S/d$	capacitance definition and a parallel-plate capacitor formula.

DIAGRAM §21

$$E = F/q$$



KEY POINT
A volt is a joule per coulomb: 1 V = 1 J/C.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 15, item 1

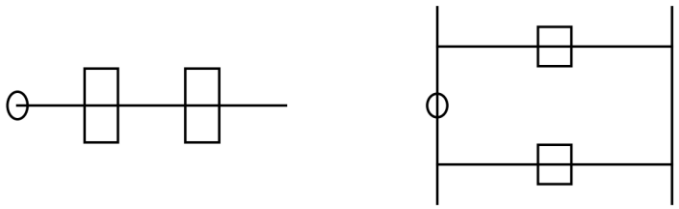
Find the potential at a point to which a charge 5×10^{-7} C is moved from infinity by doing 0.05 J of work.

Given	SI	Solution
A = 0.05 J q = 5×10^{-7} C	*All values are in SI*	$\varphi = A/q$ $\varphi = 0.05/(5\times10^{-7})$ $\varphi = 1.0\times10^{5}$ V

Answer: φ = 100 kV.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

Chapter 5. Direct electric current



§ 22. Electric current and sources of electric current

- Electric current is ordered motion of charged particles. In metals the charge carriers are free electrons; in solutions they are ions.
- A steady current requires mobile charge carriers, an electric field and a closed circuit maintained by a source.
- A source separates charges and maintains a potential difference by converting chemical, mechanical, light or other energy into electrical energy.

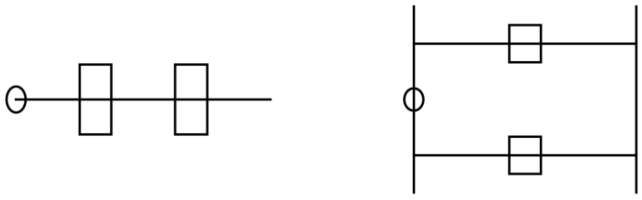
DEFINITIONS

Term	Definition
Electric current	ordered motion of charged particles.
Current source	a device that separates charges and maintains voltage in a circuit.
Conventional current direction	direction from positive to negative terminal in the external circuit, opposite to electron drift in metals.

FORMULAE AND RELATIONS

Formula	Note
$I = \Delta q / \Delta t$	current is charge passing through a cross-section per time.
$[I] = A = C/s$	one ampere is one coulomb per second.

DIAGRAM §22



KEY POINT
Do not confuse random thermal electron motion with current: only directed drift produces a net current.

§ 23. Electric circuit and its components; current and voltage

- A simple circuit contains a source, connecting wires, a load (consumer), a switch and often measuring instruments.
- An ammeter measures current and is connected in series; a voltmeter measures voltage and is connected in parallel with the component.
- Current is charge per time; voltage is work per charge. Correct polarity and suitable measurement ranges are important.

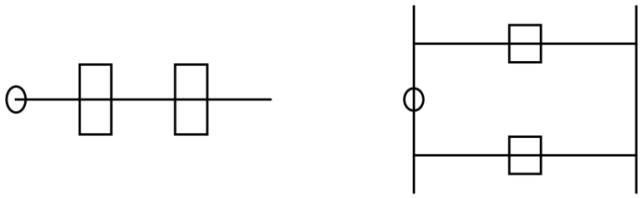
DEFINITIONS

Term	Definition
Electric circuit	a set of connected devices through which current can flow.
Ammeter	instrument that measures current; connected in series.
Voltmeter	instrument that measures voltage; connected in parallel.

FORMULAE AND RELATIONS

Formula	Note
I = q/t	current strength.
U = A/q	voltage as work per charge.

DIAGRAM §23



KEY POINT
An open switch breaks the circuit, so no steady current flows through the load.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 16, item 3

Find the current in a lamp if a charge of 300 C passes through its filament in 10 minutes.

Given	SI	Solution
q = 300 C t = 10 min	*t = 600 s*	I = q/t I = 300/600 I = 0.5 A

Answer: I = 0.5 A.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 24. Ohm's law for a section of a circuit

- For a metallic conductor at constant temperature, current is directly proportional to voltage and inversely proportional to resistance.
- The current–voltage graph of an ohmic conductor is a straight line through the origin; its slope is 1/R when I is plotted against U.

- Resistance limits current and helps determine safe operation of electrical devices.

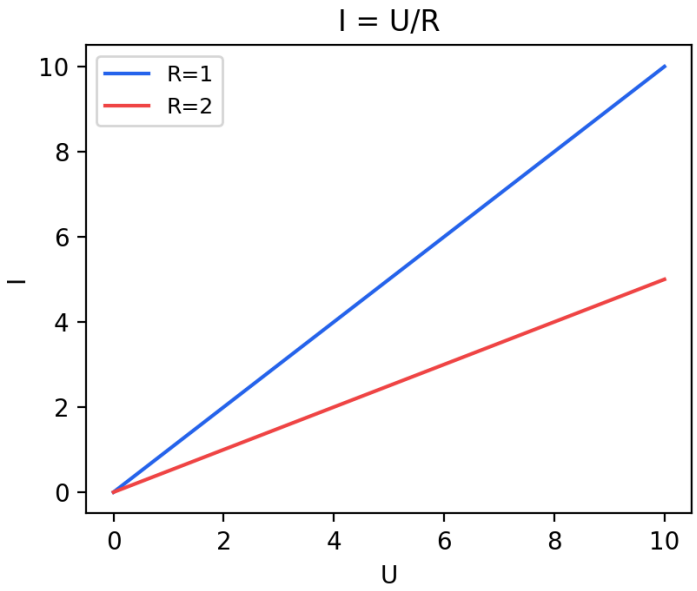
DEFINITIONS

Term	Definition
Resistance R	a property of a conductor that opposes electric current.
Ohmic conductor	a conductor obeying Ohm's law at constant physical conditions, particularly temperature.

FORMULAE AND RELATIONS

Formula	Note
I = U/R	Ohm's law for a circuit section.
U = I·R ; R = U/I	rearranged forms.
[R] = Ω = V/A	one ohm equals one volt per ampere.

DIAGRAM §24



KEY POINT
The textbook safety reminder: potentially life-threatening voltage is about 42 V in dry rooms and 12 V in wet rooms; currents above 0.05 A are dangerous.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 17, item 2

Find the voltage across a conductor of resistance 30 Ω carrying a current of 0.6 A.

Given	SI	Solution
R = 30 Ω I = 0.6 A	*All values are in SI*	U = I·R U = 0.6·30 U = 18 V

Answer: U = 18 V.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 25. Electrical resistance of a conductor, resistivity and rheostat

- The resistance of a uniform wire grows with its length and falls as cross-sectional area increases.
- Resistivity is a material property; it allows comparison of wires of the same dimensions made from different substances.
- A rheostat is a variable resistor. Changing the active length of resistance wire changes the current in a circuit.

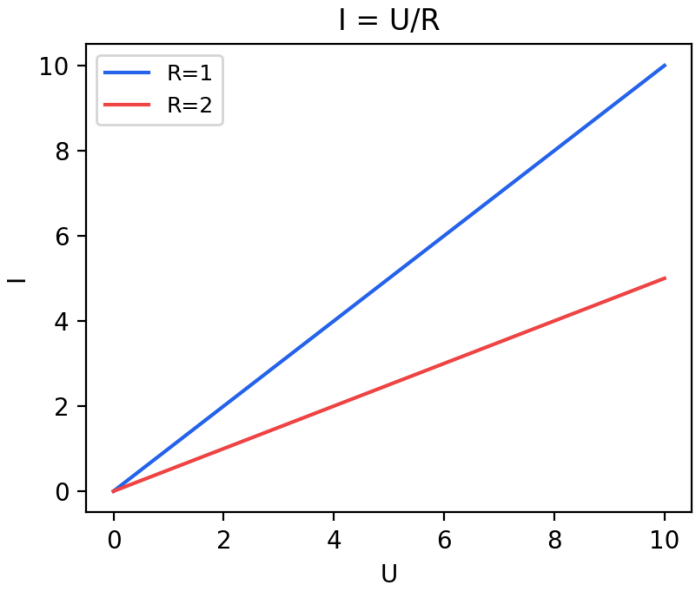
DEFINITIONS

Term	Definition
Resistivity ρ	material coefficient equal to the resistance of a conductor 1 m long with cross-section 1 m².
Rheostat	a variable resistor used to regulate current or voltage in a circuit.

FORMULAE AND RELATIONS

Formula	Note
R = ρ·l/S	resistance of a uniform conductor of length l and cross-section S.
ρ = R·S/l	definition form for determining resistivity.

DIAGRAM §25



KEY POINT
Use consistent SI units: Ω·m for resistivity, m for length and m² for cross-sectional area.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 18, item 1

Find the resistance and mass of a copper wire with cross-section 0.03 mm² and length 200 m.

Given	SI	Solution
S = 0.03 mm² l = 200 m	$S = 3\times10^{-8}\ m^2$	$R = \rho l/S = 1.68\times10^{-8}\cdot200/(3\times10^{-8}) = 112\ \Omega$ $V = Sl = 3\times10^{-8}\cdot200 = 6\times10^{-6}\ m^3$

Given	SI	Solution
$\rho_1 = 1.68\times10^{-8}\ \Omega\cdot m$ $\rho_2 = 8900\ kg/m^3$		$m = \rho_2 V = 8900\cdot6\times10^{-6} = 0.0534\ kg$

Answer: R = 112 Ω; m = 0.0534 kg.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 26. Series and parallel connection of conductors

- In a series circuit the same current flows through every component; voltages add and resistances add.
- In a parallel circuit the voltage is the same across each branch; branch currents add and equivalent resistance is less than the smallest branch resistance.
- Domestic electrical devices are connected in parallel so that each receives the supply voltage and can operate independently.

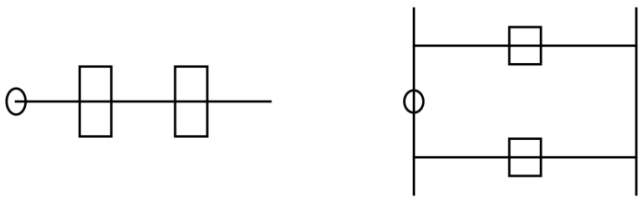
DEFINITIONS

Term	Definition
Series connection	connection in which components follow one path for current.
Parallel connection	connection in which components are connected across the same two points and form separate branches.

FORMULAE AND RELATIONS

Formula	Note
$\mathbf{I = I_1 = I_2 = \dots\ ;\ U = U_1 + U_2 + \dots\ ;\ R = R_1 + R_2 + \dots}$	laws for series connection.
$\mathbf{U = U_1 = U_2 = \dots\ ;\ I = I_1 + I_2 + \dots\ ;\ 1/R = 1/R_1 + 1/R_2 + \dots}$	laws for parallel connection.

DIAGRAM §26



KEY POINT
Before calculating a network, draw a clean circuit and decide whether each quantity is common to all components or is added across branches.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 19, item 1

Three resistors 12 Ω, 9 Ω and 3 Ω are connected in series to a 120 V supply. Find current and voltage across each resistor.

Given	SI	Solution
U = 120 V	*R = 24 Ω*	$R = R_1+R_2+R_3 = 24\ \Omega$

Given	SI	Solution
$R_1 = 12\ \Omega$ $R_2 = 9\ \Omega$ $R_3 = 3\ \Omega$		$I = U/R = 120/24 = 5$ A $U_1 = IR_1 = 60$ V; $U_2 = 45$ V; $U_3 = 15$ V

Answer: $I = 5$ A; $U_1 = 60$ V, $U_2 = 45$ V, $U_3 = 15$ V.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 27. Work and power of electric current

- Electric current does work when it transfers energy to a device. Electrical energy may become light, heat, motion or chemical energy.
- Electrical power is the rate at which electrical work is done or energy is transferred.
- Electric meters measure energy consumption, commonly in kilowatt-hours; electrical power is usually given in watts or kilowatts.

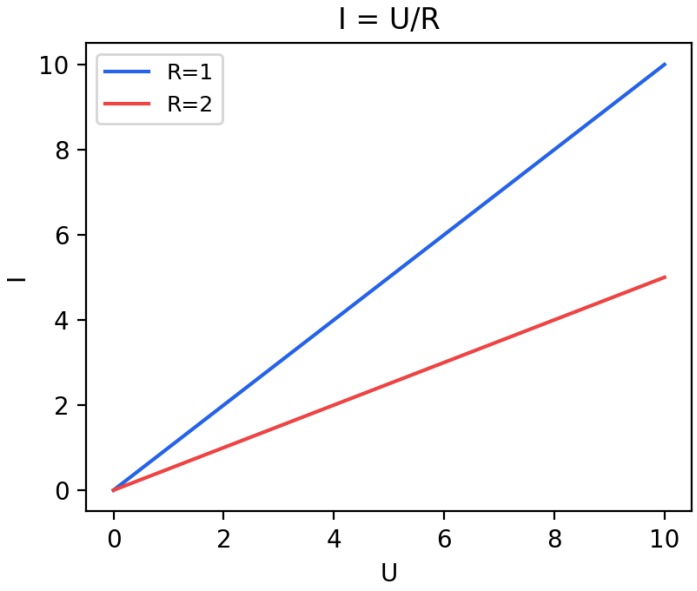
DEFINITIONS

Term	Definition
Work of current A	electrical energy transferred by current in a circuit section.
Electric power P	work of electric current per unit time.
Kilowatt-hour	energy used by a 1 kW device in one hour; 1 kWh = 3.6 MJ.

FORMULAE AND RELATIONS

Formula	Note
A = U·I·t = U·q	work of electric current.
P = A/t = U·I	power of electric current.

DIAGRAM §27



KEY POINT

Power and energy are different: a high-power device uses energy quickly, but total energy also depends on operating time.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 20, item 1

A lamp is connected to 110 V. What charge passes through its filament if the current's work is 220 J?

Given	SI	Solution
U = 110 V A = 220 J	*All values are in SI*	A = Uq q = A/U q = 220/110 = 2 C

Answer: q = 2 C.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 28. Thermal effect of electric current and the Joule–Lenz law

- Moving charges collide with particles of a conductor, so electrical energy is transformed into internal energy and the conductor heats up.
- The Joule–Lenz law gives the heat released by current in a conductor. It is especially important for heaters, fuses and preventing overheating.
- At equal voltage in parallel branches, a smaller resistance releases more power; at equal current in series, a larger resistance releases more power.

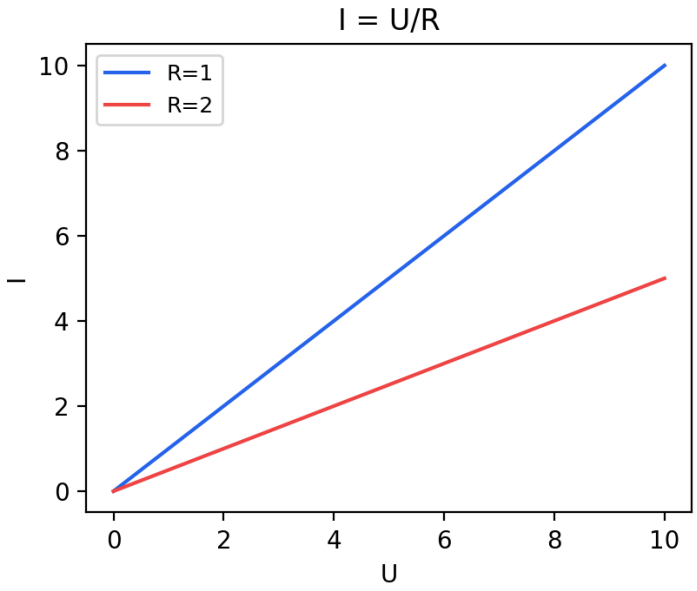
DEFINITIONS

Term	Definition
Thermal effect of current	heating of a conductor caused by electrical current.
Joule–Lenz law	law giving heat released by current in a conductor.
Fuse	a safety device whose low-melting insert opens a circuit when current becomes too large.

FORMULAE AND RELATIONS

Formula	Note
$Q = I^2 \cdot R \cdot t$	Joule–Lenz law.
$Q = U \cdot I \cdot t = U^2 t / R$	equivalent forms when using voltage and resistance.

DIAGRAM §28



KEY POINT
Never replace a fuse with a wire: a fuse is designed to interrupt a dangerous current before the wiring overheats.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 21, item 1

Two resistors 3 Ω and 6 Ω are connected in parallel. Current in the first is 2 A. What heat is released by both in 10 s, and which releases more?

Given	SI	Solution
$R_1 = 3\ \Omega$ $R_2 = 6\ \Omega$ $I_1 = 2$ A $t = 10$ s	*$U = I_1R_1 = 6$ V; $I_2 = U/R_2 = 1$ A*	$Q_1 = I_1^2R_1t = 2^2 \cdot 3 \cdot 10 = 120$ J $Q_2 = I_2^2R_2t = 1^2 \cdot 6 \cdot 10 = 60$ J $Q = Q_1+Q_2 = 180$ J

Answer: Q = 180 J; the 3 Ω resistor releases twice as much.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 29. Dependence of metal resistance on temperature; superconductivity

- For most metals, resistance increases as temperature rises because lattice vibrations hinder electron drift.
- Over a limited range, the resistance of a metal is approximated linearly using the temperature coefficient of resistance.
- Some materials become superconductors below a critical temperature: their electrical resistance becomes extremely close to zero.

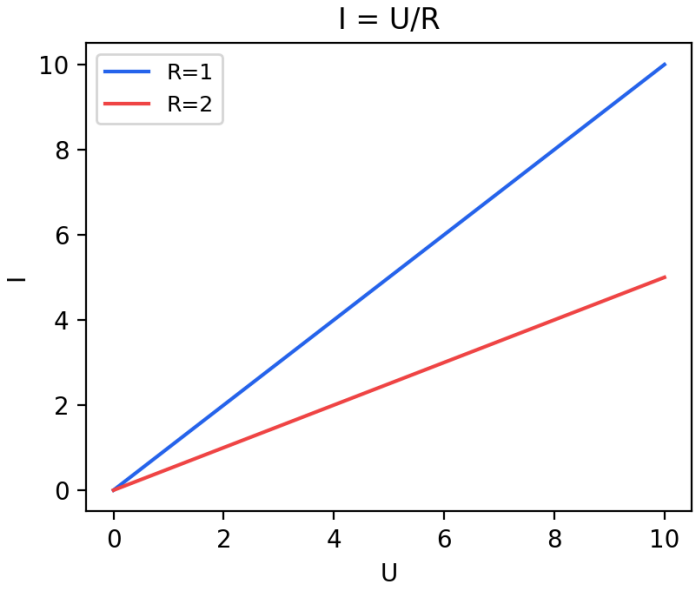
DEFINITIONS

Term	Definition
Temperature coefficient α	relative change in resistance per degree of temperature change.
Superconductivity	state of certain materials with nearly zero resistance below a critical temperature.

FORMULAE AND RELATIONS

Formula	Note
$R = R_0(1 + \alpha \cdot t)$	approximation for metal resistance at temperature t relative to 0 °C.
$\alpha = (R - R_0)/(R_0 t)$	temperature coefficient from measurements.

DIAGRAM §29



KEY POINT
The formula is a model for a stated temperature range; use α from the relevant table and keep temperature differences in kelvins or Celsius degrees.

WORKED TEXTBOOK PROBLEM — Textbook Home Exercise 22, item 1

A copper wire has resistance 4 Ω at 0 °C. Find its resistance at 50 °C if $\alpha = 4.3\times10^{-3}\ K^{-1}$.

Given	SI	Solution
$R_0 = 4\ \Omega$ $t = 50\ °C$ $\alpha = 4.3\times10^{-3}\ K^{-1}$	*All values are compatible*	$R = R_0(1+\alpha t)$ $R = 4(1+4.3\times10^{-3}\cdot50)$ $R = 4.86\ \Omega$

Answer: R = 4.86 Ω.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 30. Electric heating appliances, incandescent lamp, short circuit and fuses

- Electric heaters use the thermal effect of current. Their elements are made of high-resistivity, heat-resistant alloys such as nichrome.
- An incandescent lamp produces light when a tungsten filament is heated to a very high temperature in a bulb with little oxygen.
- A short circuit is an unintended very-low-resistance path that causes a large current. Fuses and circuit breakers protect wiring.

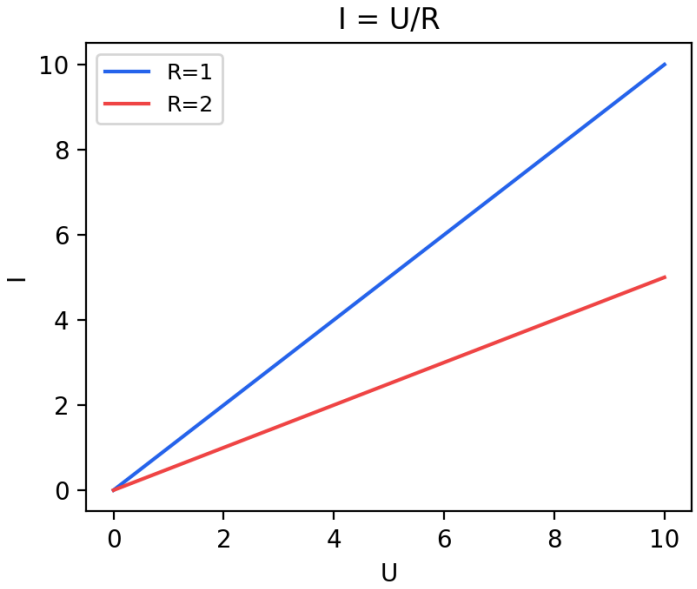
DEFINITIONS

Term	Definition
Heating element	conducting element designed to convert electrical energy into heat.
Short circuit	connection of two points of a circuit through an unintended low-resistance path.
Fuse	protective element that melts and opens a circuit when current exceeds its rated value.

FORMULAE AND RELATIONS

Formula	Note
$P = I^2R = U^2/R$	heating power of a resistor.
$Q = I^2Rt$	heat generated over time t.

DIAGRAM §30



KEY POINT
A fuse must be connected in series with the protected circuit so that it can interrupt the whole current.

§ 31. Chemical effect of electric current and Faraday's law

- Electric current through electrolytes causes chemical changes: substances can be deposited on electrodes or gases released. This is electrolysis.
- During electrolysis, positive ions move to the cathode and negative ions to the anode; reactions on electrodes form products.
- The mass of a substance deposited is proportional to the charge that passes through the electrolyte.

DEFINITIONS

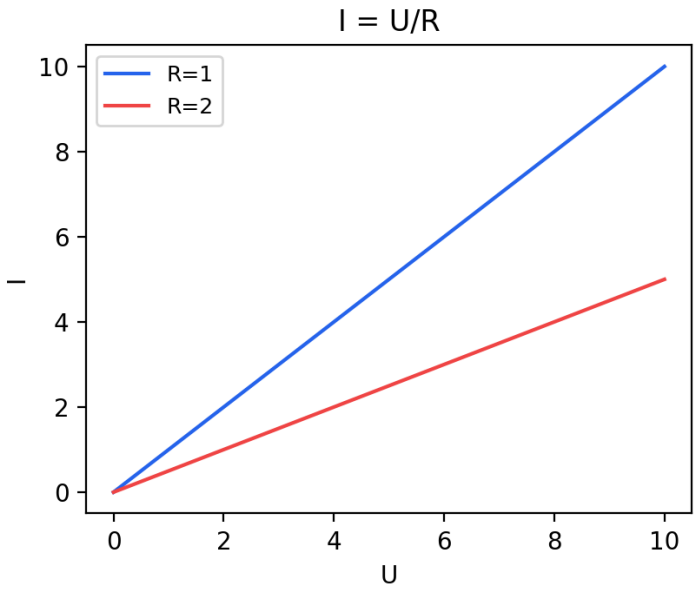
Term	Definition
Electrolysis	chemical transformations caused by electric current in an electrolyte.

Term	Definition
Electrochemical equivalent k	mass of a substance deposited by passage of unit electric charge.
Electroplating	coating an object with a metal by electrolysis.

FORMULAE AND RELATIONS

Formula	Note
m = k·q	Faraday's law in terms of charge q.
m = k·I·t	Faraday's law for constant current.

DIAGRAM §31



KEY POINT
Use seconds in m = kIt when k is in kg/C, and ensure mass units match the requested answer.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 23, item 1

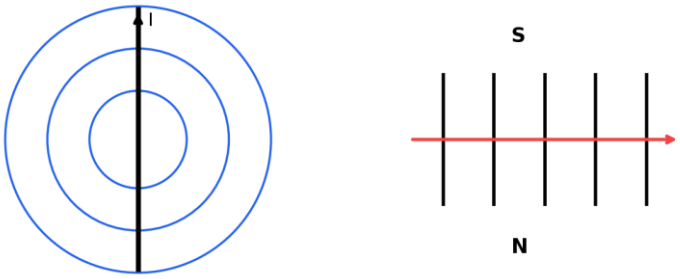
How many grams of copper are deposited from copper sulfate solution in 5 h at a current of 2 A?

Given	SI	Solution
I = 2 A t = 5 h k = 3.29×10⁻⁷ kg/C	*t = 18 000 s*	m = kIt m = 3.29×10⁻⁷·2·18 000 m = 0.011844 kg ≈ 12 g

Answer: m ≈ 12 g.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

Chapter 6. Electromagnetic phenomena



§ 32. Permanent magnets and magnetic field

- Permanent magnets have two poles, conventionally north (N) and south (S). Like poles repel and unlike poles attract.
- A magnetic field acts on magnets, moving charges and current-carrying conductors. Iron filings reveal its field-line pattern.
- Ferromagnetic materials can become magnetized because microscopic magnetic domains align. Heating or mechanical shocks can reduce magnetization.

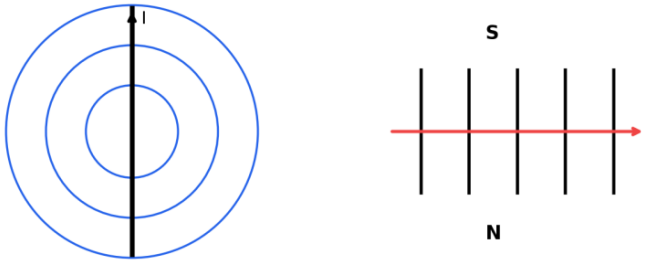
DEFINITIONS

Term	Definition
Permanent magnet	a body that retains magnetic properties for a long time.
Magnetic pole	region of a magnet where magnetic action is strongest.
Magnetic field	a form of matter responsible for magnetic interaction.

FORMULAE AND RELATIONS

Formula	Note
N ↔ S: attraction	unlike poles attract.
N ↔ N ; S ↔ S: repulsion	like poles repel.

DIAGRAM §32



KEY POINT
You cannot isolate one magnetic pole by cutting a magnet; each piece becomes a smaller two-pole magnet.

§ 33. Magnetic field of a straight current and of a current-carrying coil

- An electric current produces a magnetic field around a conductor. Around a straight wire, field lines are concentric circles.
- The direction of the field around a straight current is found by the right-hand grip rule: thumb points along conventional current, curled fingers show field direction.
- A coil produces a field similar to that of a bar magnet. Increasing current or number of turns strengthens its field.

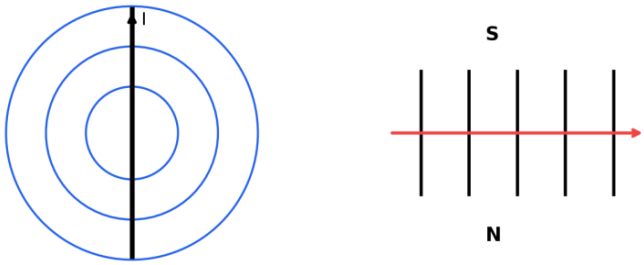
DEFINITIONS

Term	Definition
Magnetic field line	imaginary line whose tangent gives the direction of magnetic field at each point.
Solenoid / coil	a conductor wound into turns; with current it creates a magnetic field.
Right-hand grip rule	rule for finding direction of field around a current-carrying conductor.

FORMULAE AND RELATIONS

Formula	Note
I ↑ or N_turns ↑ → B ↑	qualitative dependence of coil field strength.

DIAGRAM §33



KEY POINT
The field direction inside a coil runs from its south pole toward its north pole; outside it closes back from north to south.

§ 34. Electromagnets and their applications

- An electromagnet is a coil carrying current, often with a soft-iron core. Its magnetic effect appears when current flows and can be switched off.
- Its strength increases with current, number of turns and a suitable ferromagnetic core.
- Electromagnets are used in relays, bells, lifting cranes, circuit breakers, loudspeakers and many automatic-control devices.

DEFINITIONS

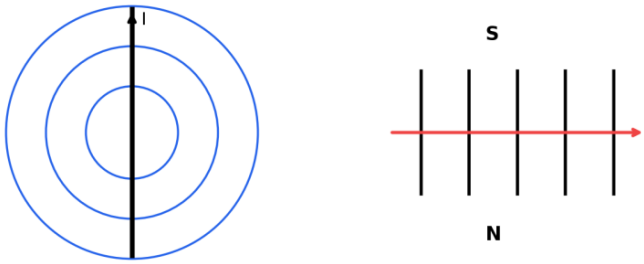
Term	Definition
Electromagnet	a magnet whose field is produced by electric current in a coil.

Term	Definition
Relay	an electromagnetic switch in which a small current controls another circuit.
Soft iron core	ferromagnetic core that is easily magnetized and demagnetized.

FORMULAE AND RELATIONS

Formula	Note
electromagnet: I = 0 → B ≈ 0	the controllable character of an electromagnet.

DIAGRAM §34



KEY POINT
To reverse the poles of an electromagnet, reverse the direction of current in its coil.

§ 35. Action of a magnetic field on a current-carrying conductor; electric motor and electrical measuring instruments

- A magnetic field exerts the Ampère force on a current-carrying conductor. Its direction depends on the directions of current and magnetic field.
- A rectangular coil in a magnetic field experiences a pair of forces, producing a turning effect. This is the basis of a DC electric motor.
- Magnetoelectric instruments use the rotation of a current-carrying coil in a magnetic field to move a pointer.

DEFINITIONS

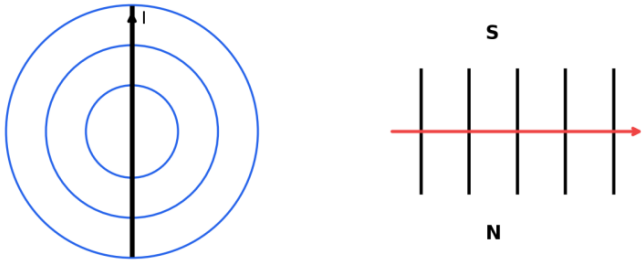
Term	Definition
Ampère force	force exerted by a magnetic field on a current-carrying conductor.
Electric motor	device that converts electrical energy into mechanical rotation.
Left-hand rule	rule for determining the direction of Ampère force.

FORMULAE AND RELATIONS

Formula	Note
$F_a \propto B \cdot I \cdot l$	magnitude of Ampère force also depends on field strength, current and conductor length in the field.

Formula	Note
electrical energy → mechanical energy	main transformation in an electric motor.

DIAGRAM §35



KEY POINT
For the left-hand rule, orient the field lines into the palm, four fingers along conventional current; the extended thumb gives the force direction.

§ 36. Electromagnetic induction and generator

- An induced current appears in a closed circuit when the magnetic field through it changes. This is electromagnetic induction.
- Moving a magnet relative to a coil, moving a coil in a field or changing the field strength can induce an emf and current.
- An electric generator converts mechanical energy into electrical energy by electromagnetic induction.

DEFINITIONS

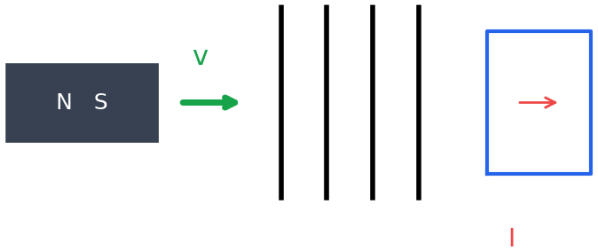
Term	Definition
Electromagnetic induction	appearance of induced emf/current in a circuit due to changing magnetic field through it.
Induced current	current caused by a changing magnetic field.
Generator	device converting mechanical energy into electrical energy.

FORMULAE AND RELATIONS

Formula	Note
$\Delta\Phi/\Delta t \neq 0$	a changing magnetic flux produces induced emf.
mechanical energy → electrical energy	energy transformation in a generator.

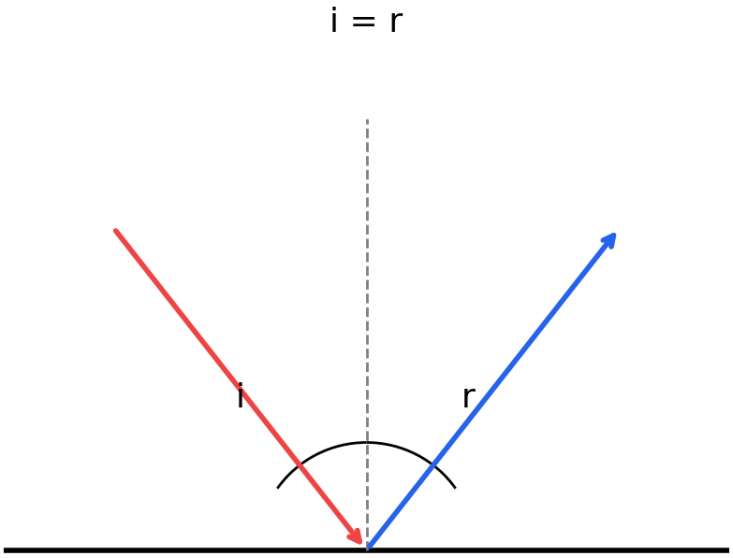
DIAGRAM §36

$$\Delta\Phi/\Delta t \neq 0$$



KEY POINT
No induced current appears while a magnet and coil remain motionless relative to each other; it is the change that matters.

Chapter 7. Light phenomena



§ 37. Law of rectilinear propagation of light

- In a homogeneous transparent medium, light travels in straight lines. Rays are a model for showing the direction of propagation.
- Opaque objects block light and form shadows; an extended source produces both a full shadow and a penumbra.
- Similar triangles in ray diagrams allow heights and distances to be found from shadow lengths.

DEFINITIONS

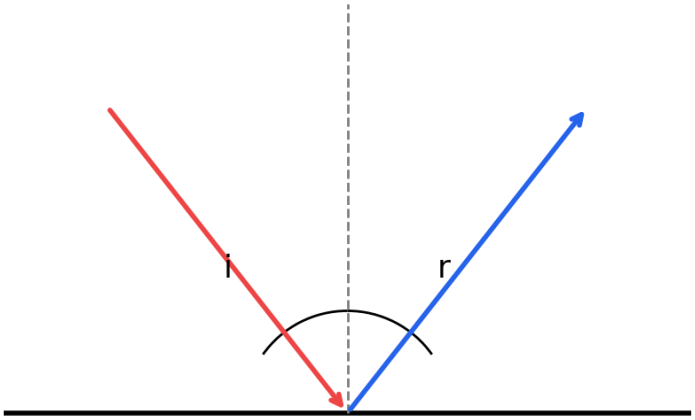
Term	Definition
Light ray	an imaginary line showing the direction of light propagation.
Shadow	region where light from a source is blocked by an opaque object.
Penumbra	partly illuminated region produced by an extended source.

FORMULAE AND RELATIONS

Formula	Note
$h_1/l_1 = h_2/l_2$	similar-triangle relation for a point source and shadows.

DIAGRAM §37

i = r



KEY POINT
Several lamps on a stadium can produce several shadows because each lamp is a separate light source.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 28, item 2

On a sunny day a person 1.8 m tall casts a 0.90 m shadow, while a tree casts a 10 m shadow. Find the height of the tree.

Given	SI	Solution
$h_1 = 1.8$ m $l_1 = 0.90$ m $l_2 = 10$ m	*Similar triangles*	$h_1/l_1 = h_2/l_2$ $h_2 = h_1 l_2/l_1$ $h_2 = 1.8 \cdot 10/0.90 = 20$ m

Answer: The tree is 20 m high.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 38. Reflection of light, laws of reflection and plane mirrors

- Reflection occurs when light changes direction at a boundary and returns to the original medium.
- The incident ray, reflected ray and normal lie in one plane; the angle of incidence equals the angle of reflection.
- A plane mirror produces a virtual, upright image of equal size at the same distance behind the mirror as the object is in front.

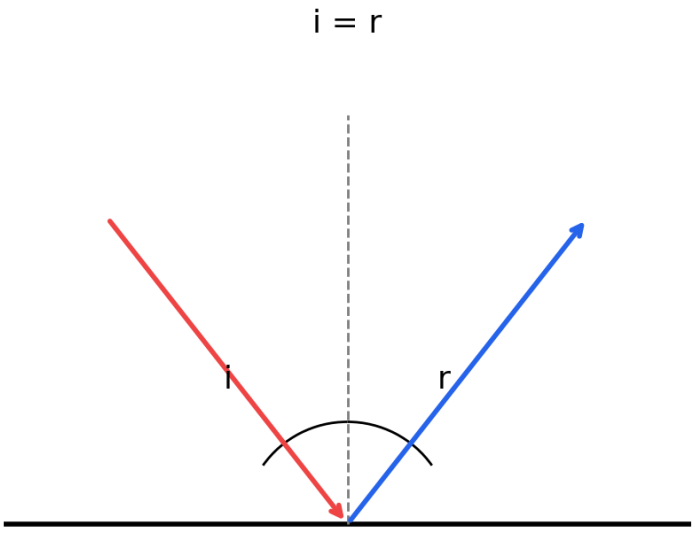
DEFINITIONS

Term	Definition
Angle of incidence i	angle between the incident ray and the normal to the surface.
Angle of reflection r	angle between reflected ray and normal.
Virtual image	image formed by apparent intersection of rays; it cannot be projected on a screen.

FORMULAE AND RELATIONS

Formula	Note
i = r	first and second laws of reflection in angular form.
d_object = d_image	for a plane mirror, measured perpendicular to the mirror.

DIAGRAM §38



KEY POINT
Angles are measured from the normal, not from the mirror surface.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 29, item 1

At what angle of incidence is the angle between the incident and reflected rays 60°?

Given	SI	Solution
φ = 60°	*All angles are in degrees*	i = r φ = i+r = 2i 2i = 60°; i = 30°

Answer: i = 30°.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 39. Spherical mirrors and image construction in a spherical mirror

- A spherical mirror is part of a spherical reflecting surface. Concave mirrors converge parallel rays; convex mirrors diverge them.
- The principal axis passes through the pole and centre of curvature. For a spherical mirror, focal length is half the radius of curvature.
- Ray construction uses principal rays: a ray parallel to the axis reflects through the focus; a ray through the focus reflects parallel to the axis.

DEFINITIONS

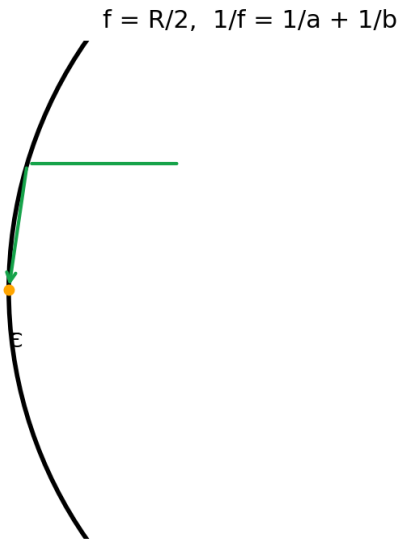
Term	Definition
Concave mirror	spherical mirror with reflecting surface on the inner side.
Convex mirror	spherical mirror with reflecting surface on the outer side.

Term	Definition
Focus F	point where parallel rays after reflection meet or appear to diverge from.

FORMULAE AND RELATIONS

Formula	Note
f = R/2	focal length f and radius of curvature R.
1/f = 1/a + 1/b	mirror formula for object distance a and image distance b, with stated sign convention.
M = |b/a| = |h'/h|	linear magnification.

DIAGRAM §39



KEY POINT
A convex rear-view mirror gives a smaller upright virtual image but a wider field of view.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 30, item 1

A body is placed 12 cm from a concave mirror with focal length 4 cm. Find image distance and the decrease in size.

Given	SI	Solution
f = 4 см a = 12 см	*All lengths are in cm*	1/f = 1/a + 1/b 1/4 = 1/12 + 1/b → b = 6 см M = b/a = 6/12 = 0.5

Answer: b = 6 cm; the image is reduced twofold.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 40. Refraction of light, law of refraction and total internal reflection

- Refraction is the change of direction of light at a boundary between transparent media because its speed changes.
- The incident ray, refracted ray and normal lie in one plane. The ratio of sines of the angles is fixed for a pair of media.

- When light travels from a denser optical medium to a less dense one, at a sufficiently large incidence angle it can undergo total internal reflection.

DEFINITIONS

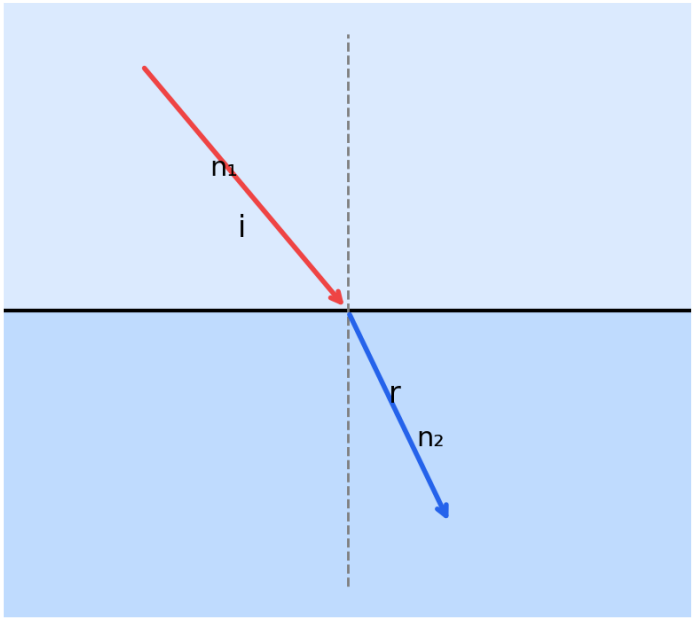
Term	Definition
Refraction	change in direction of a light ray when crossing a boundary of two transparent media.
Refractive index n	ratio c/v of light speed in vacuum to speed in the medium.
Total internal reflection	complete reflection at a boundary when light travels from optically denser to rarer medium above the critical angle.

FORMULAE AND RELATIONS

Formula	Note
$n = c/v$	definition of refractive index.
$n_1 \sin i = n_2 \sin r$	law of refraction (Snell's law).

DIAGRAM §40

$n_1 \cdot \sin i = n_2 \cdot \sin r$



KEY POINT
A spoon in water looks "broken" at the surface because rays from the submerged part are refracted on leaving water.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 31, item 2

A diver under water sees sunlight at 60° to the water surface. What is the angular height of the Sun above the horizon?

Given	SI	Solution
$\beta = 60°$ $n = 1.33$	*r = 30° to the normal*	$\sin i = n \sin r$ $\sin i = 1.33 \cdot \sin 30° = 0.665$ $i \approx 42°$; $h = 90° - i \approx 48°$

Answer: h ≈ 48°.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 41. Lenses, optical power and thin-lens formula; image construction in lenses

- A lens is a transparent body bounded by two refracting surfaces. A converging lens brings parallel rays together; a diverging lens makes them spread out.
- The focal length specifies lens action. Optical power is reciprocal focal length in metres and is measured in dioptres.
- Thin-lens ray construction uses rays parallel to the axis, passing through an optical centre, and passing through/aimed at a focus.

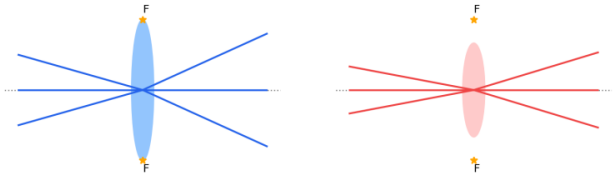
DEFINITIONS

Term	Definition
Converging lens	lens that converts a parallel beam into a converging beam.
Diverging lens	lens that converts a parallel beam into a diverging beam.
Optical power D	reciprocal of focal length in metres; unit is dioptre.

FORMULAE AND RELATIONS

Formula	Note
D = 1/f	optical power in dioptres when f is in metres; converging f and D are positive.
1/f = 1/a + 1/b	thin-lens formula.
M = |b/a| = |h'/h|	linear magnification.

DIAGRAM §41



KEY POINT
Use metres in D = 1/f. A lens with D = +1.5 dpt has f ≈ +0.67 m; a lens with D = −2 dpt has f = −0.50 m.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 32, item 1

A converging lens has optical power +1.5 dpt and a diverging lens −2 dpt. Find their focal lengths.

Given	SI	Solution
D_1 = +1.5 dpt D_2 = −2 dpt	*f is in metres*	$f_1 = 1/D_1 = 1/1.5 \approx 0.67$ m $f_2 = 1/D_2 = 1/(-2) = -0.50$ m

Answer: $f_1 \approx +0.67$ m; $f_2 = -0.50$ m.

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 42. The eye as an optical system, vision defects and their correction

- The eye forms a real, inverted image on the retina. The cornea and crystalline lens refract light; accommodation changes lens curvature to focus near and far objects.
- Myopia occurs when distant-object images form in front of the retina and is corrected by a diverging lens.
- Hyperopia occurs when near-object images would form behind the retina and is corrected by a converging lens.

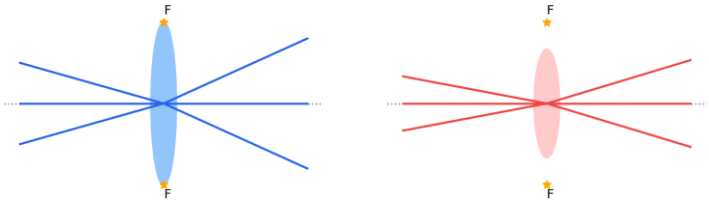
DEFINITIONS

Term	Definition
Accommodation	change of the eye lens curvature and optical power to focus objects at different distances.
Myopia	vision defect in which distant objects are focused before the retina; corrected with a diverging lens.
Hyperopia	vision defect in which near objects are focused behind the retina; corrected with a converging lens.

FORMULAE AND RELATIONS

Formula	Note
$D = 1/f$	optical power of correcting spectacles, with f in metres.
$D = 1/d_1 - 1/d_2$	for correction so that an object at normal near point is imaged at the myopic near point.

DIAGRAM §42



KEY POINT
For comfortable close work, use adequate lighting, take breaks and keep reading material at a reasonable distance.

WORKED TEXTBOOK PROBLEM — Textbook Exercise 33, item 1

Find the optical power of spectacles for a myopic person. Normal near point is 25 cm; this person's near point is 20 cm.

Given	SI	Solution
$d_1 = 0.25$ m $d_2 = 0.20$ m	*Distances are in metres*	$D = 1/d_1 - 1/d_2$ $D = 1/0.25 - 1/0.20$ $D = 4 - 5 = -1$ dpt

Answer: D = −1 dpt (a diverging lens).

Check: write quantities in SI → select the relation → substitute values with units → state the answer.

§ 43. Optical instruments

- Optical instruments use lenses, mirrors or both to collect, direct and form images of light.
- A magnifying glass gives an enlarged upright virtual image of a nearby object. A microscope uses an objective and eyepiece to view small objects.
- A telescope or binoculars helps observe distant objects by increasing their angular size; cameras project real images onto a sensor.

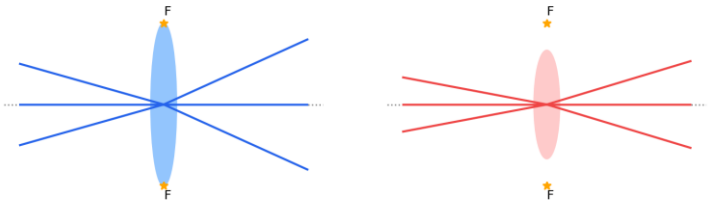
DEFINITIONS

Term	Definition
Magnifier	a converging lens used to obtain an enlarged virtual image of a close object.
Microscope	instrument for observing very small nearby objects using an objective and eyepiece.
Telescope	instrument for observing distant objects by increasing their angular size.

FORMULAE AND RELATIONS

Formula	Note
optical system → image	each instrument is analysed by the images formed by its lenses or mirrors.

DIAGRAM §43



KEY POINT
Choose an instrument by its purpose: magnification alone is not enough—brightness, resolution and field of view also matter.